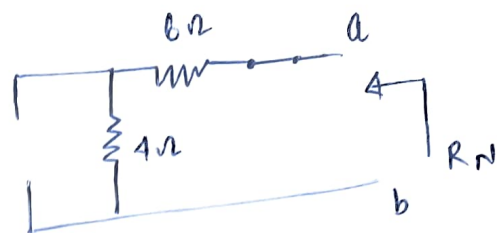


1. Ans:-

a) calculation of R_N through a-b terminal

$$R_N = (6+4)\Omega = 10\Omega$$

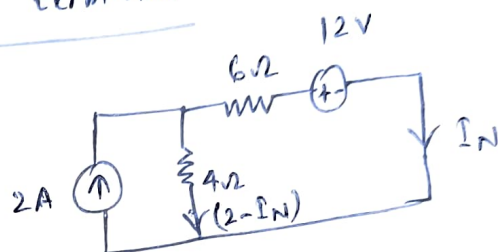
$\rightarrow (2)$



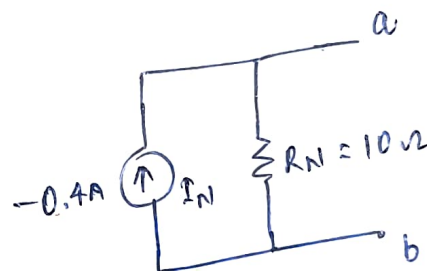
calculation of I_N through a-b terminal

$$6I_N + 12 - 4(2 - I_N) = 0$$

$$I_N = -0.4A$$



So, Norton Equivalent



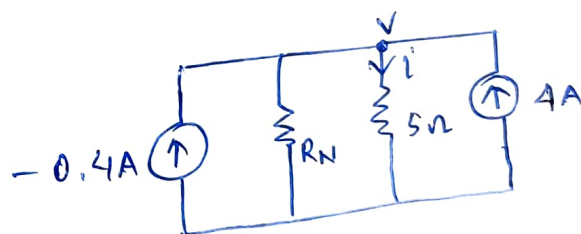
(b) apply nodal analysis at v

$$-0.4 + \frac{v}{R_N} + \frac{v}{5} - 4 = 0$$

Put $R_N = 10\Omega$ (from (a))

$$v = 12V$$

$$i = \frac{v}{5} = \frac{12}{5} = 2.4A$$



② Ans:- 6 μF & 4 μF capacitors are connected in parallel

so, $C_{eq} = (6+4) \mu\text{F} = 10 \mu\text{F}$

so, equivalent circuit will look like

Now, writing down current equations
by its waveform



$$i_s = \begin{cases} 0.02t & 0 \leq t < 1 \\ 0.02 & 1 \leq t < 2 \\ 0.01t - 0.05 & 3 \leq t < 5 \end{cases} \quad (\text{Amp})$$

Given $v(0) = 0 \text{ V}$

We know that $v_c = \frac{1}{C} \int i dt$

~~from~~ For $0 \leq t < 1$

$$v = \frac{1}{10 \times 10^{-6}} \int 0.02t dt$$

$$v = 1000t^2 + C_1$$

$$v(0) = 0 \quad ; \quad \text{so, } \boxed{C_1 = 0}$$

so, $\boxed{v = 1000t^2} \quad ; \quad 0 \leq t < 1$

For $1 \leq t < 3$

$$v = \frac{1}{10 \times 10^{-6}} \int 0.02 dt$$

$$v = (0.02 \times 10^5)t + c_2$$

as, $v(1) = 1000$ (from previous case) ; so $c_2 = -1000$

so, $v = 2000t - 1000$; $1 \leq t < 3$

For $3 \leq t < 5$

$$v = \frac{1}{10 \times 10^{-6}} \int (0.01t - 0.05) dt$$

$$v = 10^5 \left[\frac{0.01t^2}{2} - 0.05t \right] + c_3$$

as $v(3) = 5000$ (from previous case)

$$\text{so, } c_3 = 5000 - 10^5 \left[\frac{0.01 \times 9}{2} - 0.15 \right]$$

$$c_3 = 15500$$

so, $v = 500t^2 - 5000t + 15500$; from $3 \leq t < 5$

$$\text{So, } v(t) = \begin{cases} 1000t^2 & 0 \leq t < 1 \\ 2000t - 1000 & 1 \leq t < 3 \\ 500t^2 - 5000t + 15500 & 3 \leq t < 5 \end{cases} \quad (\text{Volt})$$

calculation of $i_1(t)$

$$i_1(t) = C_1 \frac{dv(t)}{dt}$$

$$\text{So, } i_1(t) = \begin{cases} 0.012t & 0 \leq t < 1 \\ 0.012 & 1 \leq t < 3 \\ 0.006t - 0.03 & 3 \leq t < 5 \end{cases} \quad (\text{Ampere})$$

③ (a) $\frac{8}{\sqrt{2}} = 5.656 \text{ A (rms)}$

(b) $f = 250 \text{ Hz}$

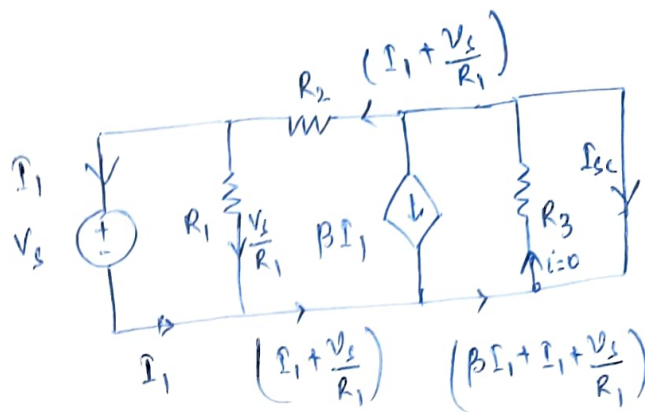
(c) $\vec{I} = \frac{8}{\sqrt{2}} \angle -25^\circ = 5.656 \angle -25^\circ \text{ A}$

(d) Both are in phase.

calculation of I_{sc}

KCL:

$$I_{sc} = - \left(\beta I_1 + I_1 + \frac{V_s}{R_1} \right) \quad \text{--- (4)}$$



KVL:-

$$V_s = -R_2 \left(I_1 + \frac{V_s}{R_1} \right) \quad \text{--- (5)}$$

From Eqⁿ (2), I_1 can be calculated as

$$I_1 = - \frac{V_s}{R_2} \left(\frac{R_1 + R_2}{R_1} \right) \quad \text{--- (6)}$$

Put I_1 expression from Eqⁿ (6) into Eqⁿ (4), we get

$$I_{sc} = 0.09965333 V_s$$

so, for max^m power transfer to R_L ;

$$R_L = R_{th}$$

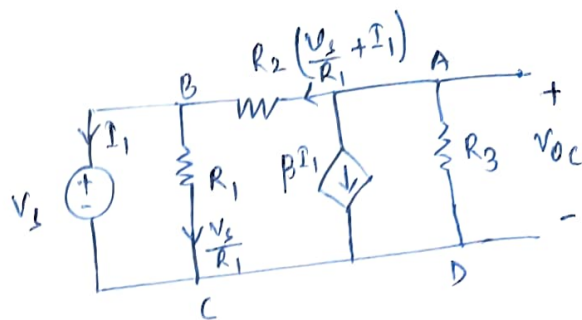
$$R_{th} = \frac{V_{oc}}{I_{sc}}$$

$$so, \quad R_L = R_{th} = 10 \Omega$$

Ans.

(4) calculation of V_{oc}

Apply nodal at V_{oc} point



$$\frac{V_{oc}}{R_3} + \beta I_1 + \frac{V_{oc} - V_s}{R_2} = 0 \quad \text{--- (1)}$$

Apply KVL in ABCD loop

$$V_{oc} = R_2 \left(\frac{V_s}{R_1} + I_1 \right) + V_s \quad \text{--- (2)}$$

From eqⁿ (2), I_1 can be calculated as

$$I_1 = \frac{R_1 (V_{oc} - V_s) - R_2 V_s}{R_1 R_2} \quad \text{--- (3)}$$

So, put the value of I_1 from eqⁿ (3) into eqⁿ (1), we get

$$V_{oc} = \left[\frac{(\beta + 1) R_1 + \beta R_2}{R_1} \right] \times \left[\frac{R_3}{R_2 + (\beta + 1) R_3} \right] \times V_s$$

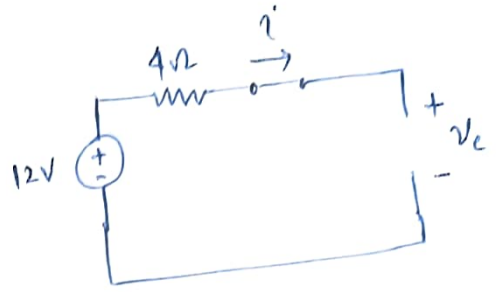
put $R_1 = 150 \text{ k}\Omega$, $R_2 = R_3 = 1 \text{ k}\Omega$ & $\beta = 98$

$$V_{oc} = \frac{3737}{3750} V_s$$

⑤ sketching the circuit before $t=0$ (when the switch was opened for a long time)

$$v_c(0^-) = v_c(0^+) = 12 \text{ V}$$

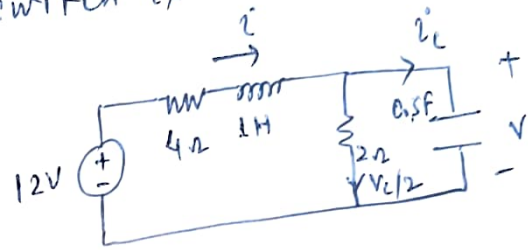
$$i(0^-) = i(0^+) = 0 \text{ A}$$



Now, sketch the circuit once the switch is closed

KCL :-

$$i = i_c + \frac{v_c}{2} \quad \text{--- (1)}$$



KVL :-

$$12 = 4i + \frac{di}{dt} + v_c \quad \text{--- (2)}$$

$$i_c = C \frac{dv_c}{dt} \quad \text{--- (3)}$$

so, (1) $\Rightarrow i = \frac{v_c}{2} + C \frac{dv_c}{dt} \quad \text{--- (4)}$

$$(2) \Rightarrow 12 = 4 \left(\frac{v_c}{2} + C \frac{dv_c}{dt} \right) + C \frac{dv_c}{dt} + v_c$$

Putting $C = 0.5 \text{ F}$ & solving we get

$$\boxed{\frac{d^2 v_c}{dt^2} + 5 \frac{dv_c}{dt} + 6 v_c = 24} \quad \text{--- (5)}$$

$Eq^n(s)$ represents 2nd order non-homogeneous 2nd order differential Eq^n . so, its complete solution will be summation of complementary solⁿ & particular integral solⁿ

so,
$$v_c(t) = v_{cc}(t) + v_{cp}(t)$$

$\Downarrow$ complete solⁿ $\Downarrow$ complementary solⁿ $\Downarrow$ p.i. solⁿ

solving complementary solⁿ

$$\frac{d^2 v_c}{dt^2} + 5 \frac{dv_c}{dt} + 6 v_c = 0$$

$$m^2 + 5m + 6 = 0$$

$$(m+2)(m+3) = 0$$

$$m = -2 \quad \& \quad -3$$

(overdamped solⁿ)

so,
$$v_{cc}(t) = Ae^{-2t} + Be^{-3t}$$

solving particular integral solⁿ

Take $v_c = Ke^{0.x}$

Now,
$$\textcircled{5} \Rightarrow 6(K e^{0.x}) = 24$$

$$K = 4$$

so,
$$v_c(t) = 4 + Ae^{-2t} + Be^{-3t}$$

— $\textcircled{6}$

Now,

using initial condition to get the values of A & B

$$v_c(0^-) = v_c(0^+) = v_c(0) = 12 \text{ V}$$

so, eqn (6) $\Rightarrow 12 = 4 + A + B$

$A + B = 8$

 — (7)

consider eqn (4)

$$i = \frac{v_c}{2} + C \frac{dv_c}{dt}$$

$$i(0) = \frac{v_c(0)}{2} + \left(\frac{dv_c(0)}{dt} \right) \times \frac{1}{2} \quad \text{--- (8)}$$

$i(0^-) = i(0^+) = 0 \text{ A}$

 $\rightarrow$ initial condition

so,

(8) $\Rightarrow \frac{dv_c(0^+)}{dt} = -12$ — (9)

so, differentiate eqn (6) to get $\frac{dv_c}{dt}$

$$\frac{dv_c}{dt} = -2Ae^{-2t} - 3Be^{-3t} \quad \text{--- (10)}$$

at $t = 0^+$, $\frac{dv_c(0)}{dt} = -12$ [from eqn (9)]

so, (10) $\Rightarrow$

$2A + 3B = 12$

 — (11)

solving eqⁿ (7) & (11)

$$\boxed{A = 12}, \quad \boxed{B = -4}$$

so, $\boxed{v_c = 4 + 12e^{-2t} - 4e^{-3t}}$ — (12) ; $t > 0$

eqⁿ (4) $\Rightarrow i = \frac{v_c}{2} + C \frac{dv_c}{dt} \Rightarrow i = 2 - 0.5Ae^{-2t} - Be^{-3t}$ — (13)

$\boxed{\dot{v} = 2 - 0.5A - B}$ — (14) (from eqⁿ 12) and

using $i(0) = 0$ (initial condⁿ)

Differentiate eqⁿ (13) to get $\frac{di}{dt}$

$$\boxed{\frac{di}{dt} = Ae^{-2t} + 3Be^{-3t}}$$
 — (15)

consider eqⁿ (2)

$$12 = 4i + \frac{di}{dt} + v_c$$

at $t=0$, $i(0) = 0$, $v_c(0) = 12$

so, $\boxed{\frac{di(0)}{dt} = 0}$

so, eqⁿ (15) $\Rightarrow$ at $t=0$, $\boxed{A + 3B = 0}$ — (16)

Now, solve eqⁿ (14) & (16)

$$\boxed{A = 12}, \quad \boxed{B = -4}$$

$$\boxed{i(t) = 2 - 6e^{-2t} + 4e^{-3t}} \quad ; \quad t > 0$$

6) Given

1. All elements are fully discharged in beginning

It means, $i_L(0^-) = i_L(0^+) = 0 \text{ A}$

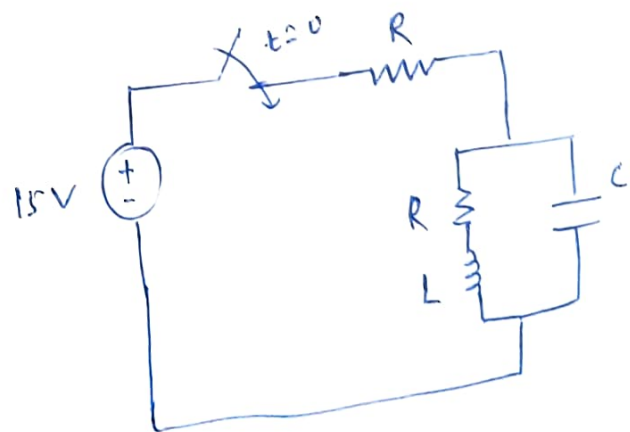
$$v_C(0^-) = v_C(0^+) = 0 \text{ V}$$

2. From the current vs time plot, $i(0)$ is higher than $i(10)$ i.e. $i(\infty)$

3. We know that inductor & capacitor behaves as short circuit & open circuit respectively in DC steady state

So, at $t = 0$, capacitor will be short ckt, because $v(0) = 0 \text{ V}$

$$\text{So, } i(0) = \frac{15}{R}$$



at $t = \infty$, L will be short, C will be open

$$\text{So, } i(\infty) = \frac{15}{2R}$$

and hence $\boxed{i(0) > i(\infty)}$ proved

⑦

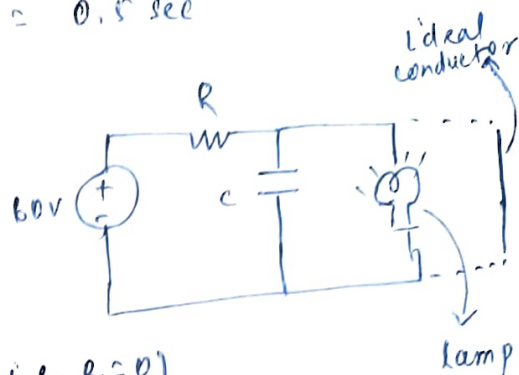
Time period of flash = $\frac{1}{2}$ sec = 0.5 sec

so, as the voltage reaches to 30V across the neon lamp, it starts

discharging through ideal conductor (i.e. $R_{ideal} = 0$).

i.e. discharging time can be taken as zero.

so, $\frac{T}{2} = 0.5$ sec for one cycle



$$V(t) = V(\infty) + [V(0) - V(\infty)] e^{-t/\tau}$$

$$30 = 60 + [0 - 60] e^{-t/\tau}$$

$$30 = 60(1 - e^{-t/\tau})$$

$$\text{put } t = 0.5$$

$$0.5 = 1 - e^{-0.5/\tau}$$

$$\tau = 0.7215$$

$$RC = 0.7215$$

$$R = \frac{0.7215}{10 \times 10^{-6}}$$

$$R = 0.7215 \times 10^5$$

$$R = 72.15 \text{ k}\Omega$$

Q. 8) Ans.

1. > Arc is generated across the switch contact due to sudden change in inductor current as inductor doesn't allow sudden change in current.
2. > Putting capacitor in series with L & R prevents sudden change in current. The capacitor will take some time to get charge as ~~it~~^{of} source of voltage value. So, voltage buildup time across capacitor will allow gradual decrease in inductor current & prevents arc across switch contact.
3. > Also, to get an immediate response means as the switch opens, current in motor field need to be zero, the response should be critically damped because critically damped response is fastest among all the damped response.

So, for critically damped response

$$\alpha = \omega_0$$

$$\frac{R}{2L} = \frac{1}{\sqrt{LC}}$$

$$\frac{R^2}{4L^2} = \frac{1}{LC}$$

$$C = \frac{4L}{R^2} = 320 \mu F$$